

Using the neutrophil-to-lymphocyte ratio to predict outcomes in pediatric patients with traumatic brain injury

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ABSTRACT

Objective: The brain's inflammatory reaction to traumatic brain injury (TBI) generally peaks between 24 and 48 h after injury. This inflammatory cascade can be neuroprotective or may mediate secondary brain injury beyond the initial TBI. Therefore, circulating inflammatory markers may be useful for predicting outcomes in pediatric TBI. The goal of this study was to determine whether elevations in peripheral blood neutrophil-to-lymphocyte ratios (NLRs) are associated with adverse outcomes in pediatric TBI patients.

Patients and Methods: 188 pediatric patients (0–18 years) presenting to our institution with TBI from 2007 to 2017 were retrospectively reviewed. Absolute neutrophil and lymphocyte counts from a complete blood count (CBC) were used to calculate NLRs on admission (< 12 h) and approximately 24, 48, and 72 h after injury. Data points included Glasgow Coma Scale (GCS) on admission, presence of post-traumatic amnesia (PTA), loss of consciousness (LOC), and Glasgow Outcome Scale Extended Pediatric Version (GOS-E Peds) with a median outcome span of 86 days.

Results: A one-way ANOVA demonstrated statistically significant differences in NLR at 24 h ($p = 0.004$) and 48 h ($p = 0.003$) among patients stratified by GOS-E Peds. No significant differences in NLR were observed at any time point based on GCS or PTA. Patients who experienced LOC had a significantly higher NLR on admission ($p = 0.013$) and at 24 h ($p < 0.001$) than those who did not.

Conclusion: In this study, relatively higher NLRs at 24 and 48 h post-TBI were associated with worse outcomes in pediatric patients. This suggests that NLR may be a useful and cost-effective outcome predictor in pediatric TBI as well as a possible future target for therapeutic intervention, warranting larger prospective trials.

1. Introduction

Traumatic brain injury (TBI) is the leading cause of death and neurologic disability in children after infancy [1]. Despite this, there is a paucity of validated clinical tools for predicting outcomes after pediatric brain injury. The majority of clinically available outcome tools are extrapolated from adult data, and, given the substantial differences between developing and mature brains, these tools are likely less accurate in pediatric patients [1,2]. Limited previous pediatric studies have found that poor outcomes are associated with the following variables: low GCS at 24 h, hypoxia on admission, CT with subarachnoid hemorrhage, diffuse axonal injury, brain swelling, and poor

control of cerebral perfusion pressure [1,3]. Currently, the most sensitive clinical tools for prognostication of pediatric TBI at the initial time of injury are assessments of gross motor function, functional independence, health-related quality of life, time to follow commands, PTA, and neuropsychological function [1,4,5]. This study used the Glasgow Outcome Scale Extended Pediatric Version (GOS-E Peds) to assess final TBI outcome, as it is a clinically accepted and validated method for reporting pediatric TBI outcomes [6,7].

After a TBI, a complex immune injury response is initiated, and white blood cells, including neutrophils, monocytes, and lymphocytes, are recruited to the damaged areas to ameliorate any threats of cell damage, eliminate necrotic cells, and encourage tissue repair. [8]

Abbreviations: AANS, American Association of Neurological Surgeons; ANOVA, analysis of variance; CBC, complete blood count; CDC, Centers for Disease Control and Prevention; CT, computed tomography; GCS, Glasgow Coma Scale; GOS-E Peds, Glasgow Outcome Scale Extended Pediatric Version; IRB, institutional review board; LOC, loss of consciousness; NLR, neutrophil-to-lymphocyte ratio; PTA, post-traumatic amnesia; TBI, traumatic brain injury

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Preclinical studies have shown that this inflammatory cascade peaks 24–72 h after injury, but the full immunopathologic time-course has yet to be characterized in humans [9–12]. There is a large body of evidence that suggests that this neuroinflammation, including the upregulation of inflammatory mediators, can provide neuroprotection after a TBI [8,13–15]. However, excessive neuroinflammation may also contribute to ongoing neurodegeneration, neurological impairment, and maladaptive secondary injury reactions [8,13,16].

Extensive preclinical research is underway to better understand the effects of the immune response on TBI pathophysiology, but there is a paucity of clinical data available, especially in the pediatric population [16,17]. This retrospective study aims to measure blood NLR and correlate it with TBI outcome scores.

2. Patients and methods

2.1. Study design

This study was approved by the Memorial Healthcare System Institutional Review Board (IRB # MHS.2017.100). A retrospective review of 188 pediatric patients presenting to our institution with TBI from January 1, 2007, to December 31, 2017, was performed using our institution's Trauma One registry (Lancet Technology, Inc.; Boston, MA). Inclusion criteria were age 0–18 years, diagnosis of an isolated TBI, and at least one complete blood cell (CBC) panel with differential taken within 84 h of the time of injury. Exclusion criteria included severe clinically diagnosed comorbidities; prior neurological disease, including ischemic or hemorrhagic stroke; use of anticoagulant medication, steroids, or immunosuppressants; and prior systemic disease including uremia, liver cirrhosis, malignancy, chronic heart disease, diabetes, and hypertension.

Individual NLRs were calculated by dividing the absolute neutrophil count by the absolute lymphocyte count. The average patient NLR was calculated for four time periods based on time after injury: < 12 h (range: 0–11.9 h post-injury), 24 h (range: 12–35.9 h post-injury), 48 h (range: 36–59.9 h post-injury), and 72 h (range: 60–84 h post-injury). When multiple data points within a time frame were present for a single patient, the CBC obtained closest to the time of interest was used. Clinical presentation data points included GCS score on admission, presence of PTA, and LOC at the time of injury. Outcomes were classified based on patient survival and Glasgow Outcome Scale Extended Pediatric Version (GOS-E Peds) score. PTA and LOC are recorded in the Trauma One Registry by trained registry personnel, and this information was taken directly from the registry. The GOS-E Peds score was determined from a chart review performed by the senior author (H.S.) for each patient based on the most recent information on functional outcome available at the time of chart review and blinded to NLR values. TBI severity was classified based on admit GCS: mild (13–15), moderate (9–12), or severe (3–8). Patient injury outcomes were stratified based on GOS-E Peds score: none to mild disability (1–2), moderate to severe disability (3–6), or vegetative state/death (7–8) [6]. If a patient did not have a particular variable documented in the Trauma One Registry or if their outcome could not be reliably scored, then they were excluded from that specific analysis. To account for having fewer patients with data available at later time points, post hoc testing was conducted for the subset of patients for whom NLR data were available at both < 12 h and 48 h.

2.2. Statistical analysis

The population mean and standard deviation were calculated for study variables at each post-injury time point. Two-sample t-tests (two-tailed) were performed for the binary variables: LOC, PTA, and patient survival. A one-way analysis of variance (ANOVA) was used to determine differences among outcome groups for GOS-E Peds and admission GCS. Post-hoc multi-comparison analysis using the Tukey-

Table 1

Demographic and Clinical Characteristics of 188 patients with TBI.

Demographic & Clinical Characteristic	N (%)
Mean Age (years) $\pm$ SD	9.49 $\pm$ 6.70
Median Length of Hospital Admission (days), Range	3, [1–48]
Median Outcome Span (days), Range	85.5, [1–1433]
Sex	
Male	118 (63 %)
Female	70 (37 %)
Race	
White	126 (67 %)
Black	57 (30 %)
Asian	4 (2%)
Other	1 (1%)
GCS Score on Admission	
Severe: 3–8	24 (13 %)
Moderate: 9–12	9 (5%)
Mild: 13–15	149 (82 %)
Loss of Consciousness	
Y	94 (52 %)
N	88 (48 %)
Post-Traumatic Amnesia	
Y	20 (22 %)
N	71 (78 %)
GOS-E Peds Score	
No/Mild Disability: 1–2	163 (88 %)
Moderate/Severe Disability: 3–6	14 (8%)
Vegetative State/Death: 7–8	9 (5%)
Survival	
Y	181 (96 %)
N	7 (4%)

Abbreviations: GCS = Glasgow Coma Scale, GOS-E Peds = Glasgow Outcome Scale Extended Pediatric Version.

Variables may not total 100 % due to rounding-up.

Kramer method was performed for any significant findings to reveal where statistically significant differences occurred between groups. The Welch statistic was used for any data that failed a test of homogeneity of variances. Post hoc testing using Spearman's rank-order test was conducted to compare NLR and GOS-E Peds for the subset of patients for whom NLR data were available at both < 12 h and 48 h. All data were analyzed using SPSS Statistics, version 25 (IBM, Chicago, IL).

3. Results

3.1. Demographics

A total of 199 records were reviewed, of which 188 patients met inclusion criteria. The mean patient age was 9.49 ± 6.70 years (median = 11.5, range = 0–18 years). The median length of hospital stay was 3 days (range = 1–48), and the median length of time to follow-up was 85.5 days (range = 1–1433). A summary of patient demographic data is presented in Table 1, and the breakdown of the specific data available for each time period is summarized in Fig. 1. The overall mean NLR was 4.30 ± 6.00 within 12 h of injury (N = 156), 5.10 ± 5.95 at 24 h (N = 58), 6.03 ± 3.59 at 48 h (N = 27), and 8.10 ± 9.98 at 72 h (N = 21).

3.2. Patient presentation and injury symptoms

No significant differences in NLR or neutrophils were observed at any time point based on GCS category (Supplemental Table S1). There were also no significant differences in NLR or neutrophil or lymphocyte absolute counts at any time point based on presence of PTA (Supplemental Table S2). Patients who experienced LOC had a significantly higher first NLR (< 12 h) on admission ($p = 0.013$) and at 24 h ($p < 0.001$) compared to those without LOC (Table 2). At 72 h, there were too few data points in one or more groups for GCS and PTA for results to be meaningful; therefore, p-values were not computed

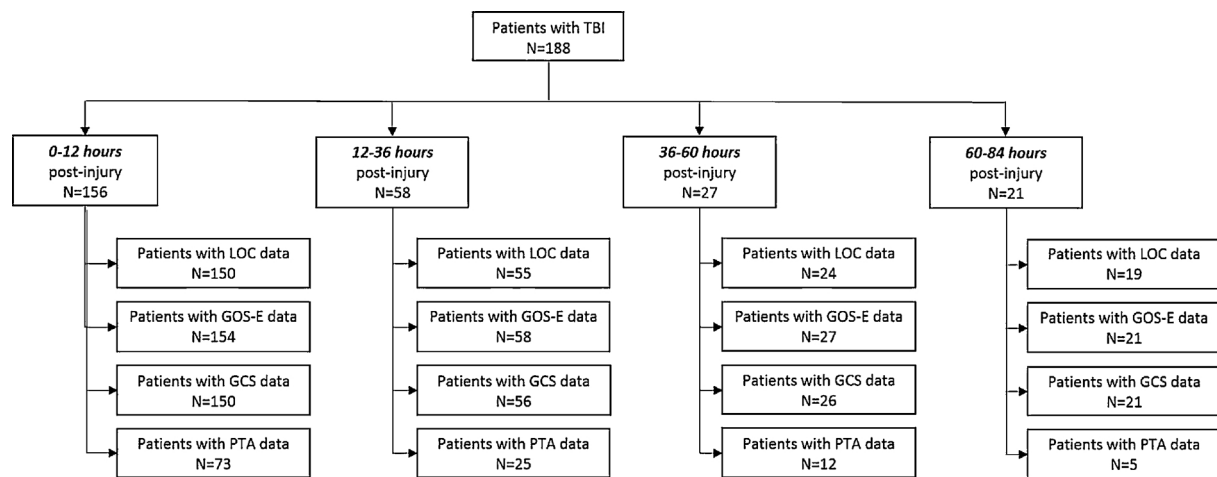


Fig. 1. Flow chart indicates data available for patients at different study time points.

(Tables S1 and S2).

3.3. Patient outcomes correlated with NLR

A one-way ANOVA demonstrated statistically significant differences in NLR at 24 h ($p = 0.004$) and 48 h ($p = 0.003$) among patients stratified by GOS-E Peds score, and the post hoc analyses using the Tukey-Kramer test determined a significant difference in NLR at 24 h between the mild and moderate ($p = 0.047$) and the mild and severe ($p = 0.018$) GOS-E Peds groups (Table 3). At 48 h, significant differences in NLR were found between the mild and severe ($p = 0.002$) and the moderate and severe ($p = 0.030$) GOS-E Peds groups. There were no significant differences in NLR among GOS-E Peds groups at time of admission (< 12 h) or at 72 h. Results of other immune cells are presented in Supplemental Table S3. Fig. 2 shows the time course of the average NLR for patients in each GOS-E Peds outcome category. The figure depicts a gradual increase, after admission, in average NLR among the “vegetative state/death” GOS-E group. Furthermore, when comparing all three GOS-E groups, at 24 h and 48 h CBC time, patients in the “vegetative state/death” and “moderate/severe disability” GOS-E Peds groups had

higher average NLRs when compared to the “no/mild disability” GOS-E Peds group. When restricted to patients for whom NLR data were available at both < 12 h and 48 h, there was a statistically significant, strong positive correlation between NLR and GOS-E Peds at 48 h [$r_s(16) = 0.594$, $p = 0.009$], such that worse outcomes were correlated with higher NLRs. Within this subset, there was no correlation between GOS-E Peds and NLR at < 12 h.

In terms of survival status, the majority of patients survived (96 %, $n = 181$) (Table 1). Of the seven patients who died, six presented with a GCS of 3. Two patient deaths occurred within the time frame observed in this study, and the remaining five occurred within the first few weeks after TBI (range = 19–420 h post-injury). A two-sample, two-tailed t -test based on patient survival revealed that patients who did not survive had a significantly higher NLR at both 24 h ($p = 0.048$) and 48 h ($p = 0.01$). Additionally, the mean absolute neutrophil count at 24 h was significantly higher among patients who did not survive ($p = 0.001$). No differences based on survival were observed at < 12 h (Supplemental Table S4). At 72 h, there were only 2 patients who had a CBC with differential available. This group size was determined to be too low for results to be meaningful; therefore, statistical tests were not

Table 2

Comparison of NLR and immune cell counts based on loss of consciousness after injury.

Time after Injury	CBC Panel Results	With LOC			Without LOC			P-Value*
		n	Mean	SD	n	Mean	SD	
< 12 Hours	NLR	81	5.23	7.02	69	2.94	3.80	0.013
	% Neutrophils		60.73	18.40		50.30	22.69	0.003
	Absolute Neutrophils		8.96	5.98		6.65	4.47	0.01
	% Lymphocytes		28.06	18.31		40.18	22.60	0.001
	Absolute Lymphocytes		3.06	1.86		4.60	2.85	< 0.001
24 Hours	NLR	35	6.81	3.92	20	2.06	1.68	< 0.001
	% Neutrophils		71.77	12.32		51.38	16.43	< 0.001
	Absolute Neutrophils		9.13	4.44		5.19	2.03	< 0.001
	% Lymphocytes		15.35	9.23		36.10	15.99	< 0.001
	Absolute Lymphocytes		1.57	0.67		3.69	2.19	< 0.001
48 Hours	NLR	20	6.24	3.69	4	3.74	3.34	0.22
	% Neutrophils		71.59	12.77		62.65	15.87	0.23
	Absolute Neutrophils		8.31	5.34		6.03	2.43	0.42
	% Lymphocytes		16.64	10.93		23.48	10.51	0.26
	Absolute Lymphocytes		1.53	0.67		2.13	1.00	0.15
72 Hours	NLR	14	9.87	11.14	5	1.49	1.67	0.12
	% Neutrophils		72.36	10.44		40.32	24.07	0.04
	Absolute Neutrophils		9.79	7.27		3.09	1.54	0.06
	% Lymphocytes		13.29	8.03		48.06	27.19	< 0.001
	Absolute Lymphocytes		1.37	0.78		4.92	4.39	0.15

Abbreviations: CBC = complete blood count; LOC = loss of consciousness; NLR = neutrophil-to-lymphocyte ratio; SD = standard deviation.

* Boldface values indicate significance at $\alpha = 0.05$.

Table 3

ANOVA comparison and Tukey-Kramer post-hoc analysis of NLR and immune cell counts based on GOS-E Peds score.

Time after Injury	GOS-E Peds Score	n	NLR		ANOVA P-Value	Post-hoc P-Value*			
			Mean	SD		GOS-E Peds 1-2	GOS-E Peds 3-6		
< 12 Hours	1-2	134	4.15	5.87	0.38				
	3-6	11	6.79	8.42					
	7-8	9	4.13	4.94					
24 Hours	1-2	46	4.25	3.43	0.004	0.047	0.83		
	3-6	7	7.84	4.27					
	7-8	5	9.08	4.55					
48 Hours	1-2	18	4.92	3.05	0.003	0.80	0.03		
	3-6	5	5.86	2.98					
	7-8	4	11.22	1.95					
72 Hours	1-2	13	7.96	12.50	0.80				
	3-6	5	6.45	3.58					
	7-8	3	11.45	2.85					

Abbreviations: GOS-E Peds = Glasgow Outcome Scale Extended Pediatric Version; NLR = neutrophil-to-lymphocyte ratio; SD = standard deviation.

* Boldface values indicate significance at $\alpha = 0.05$.

conducted for this time point.

4. Discussion

This retrospective study of pediatric patients with TBI revealed that patients with worse GOS-E Peds scores exhibited significantly higher peripheral NLRs at 24 and 48 h post-injury, while the NLRs measured within the first 12 h post-injury had no association with GOS-E Peds score. There was no significant difference in NLR among GOS-E Peds groups after post-injury time of 72 h, and there was no association between NLR and GCS or PTA at any time point. Patients with LOC had a significantly higher NLR on admission (< 12 h) and at 24 h compared with those who did not lose consciousness.

4.1. Serum markers in pediatric TBI

Although the use of serum markers is not currently recommended by the Centers for Disease Control and Prevention (CDC) because they have not shown consistent predictive ability, there are a small number of studies of severe pediatric TBI that have shown that these markers have some prognostic value [18–22]. One retrospective study of 57 children showed that children with hyperglycemia after 48 h have worse outcomes [21]. Another cross-sectional study of 108 patients found that an initial mixed metabolic acidosis plus respiratory acidosis

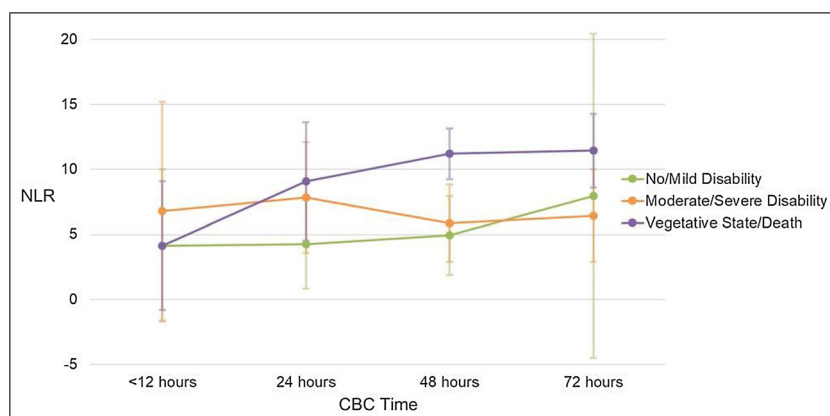
and GCS are significant predictors of mortality [20]. Due to the lack of consistency of laboratory investigations for pediatric TBI and the limited availability of these data retrospectively, this study did not analyze glucose or blood gases in pediatric TBI. There have been limited pediatric studies on the body's immune response to TBI. Hori et al. conducted a retrospective study of 22 patients which found that eosinopenia at the time of hospital admission was associated with poor outcome in pediatric patients with traumatic intracranial hemorrhage [18]. Pagowska-Klimek et al. compared 16 pediatric patients with severe TBI and found an increase in overall white blood cells with an increase in granulocytes and decrease in lymphocytes within one hour of injury compared to a non-TBI control group [22].

4.2. NLR and prognostication in neurological injuries

NLR has been studied as a predictive factor for multiple neurological conditions, including intracerebral hemorrhage, stroke, adult TBI, and pediatric brain tumors [17,23–27]. In adult patients with severe TBI (GCS < 9), Chen et al. observed that increased NLRs were associated with unfavorable 1-year functional outcomes and mortality [16]. Petrone et al. found that at 0–6 h and 24 h post-injury, NLR was higher in adults with mTBI compared with adults with severe TBI [17]. Zhao et al. developed a predictive model using traditional prognostic factors and observed that including NLR improved the model's ability to predict 6-month outcome in adult TBI [28]. In addition to its utility in evaluating neurological conditions, NLR has also proven to be useful in adults for stratifying mortality in major cardiac events, prognosticating several types of cancers, and predicting inflammatory or infectious pathologies [16,29–34].

A TBI consists of two events: (1) the primary injury, or the original force that causes damage to the neural tissue; and (2) the secondary injury, the body's inflammatory response to the injury that can cause blood-brain barrier disruption, neurotoxicity, neuronal death, and/or neurodegeneration [8,13,14]. As with any injury, TBI causes a neutrophil injury-response, which homeostatically initiates repair processes via interactions with brain resident microglia and peripheral monocyte-derived macrophages. Therefore, it is not surprising that NLR values are similar within the first 12 h after an injury. In our study, this was reflected by the lack of correlation between NLR and GOS-E Peds for the NLR obtained at the time of admission. Once the TBI progresses to critical secondary brain injury, the body's prolonged immune response can be measured by changes in neutrophils and lymphocytes.

The exact pathophysiology of the neutrophil response after a TBI has yet to be elucidated. However, there are several important roles of the neutrophil that have been defined in injury responses and in the orchestration of pro-repair mechanisms. Neutrophils perform an important role by connecting the innate and adaptive immune systems [15]. The innate immune system is made up of predominately neutrophils, which are rapidly activated after an injury and migrate from

**Fig. 2.** Line chart shows the progression of neutrophil-to-lymphocyte ratio by GOS-E Peds groups.

Time course of the average neutrophil-to-lymphocyte ratio (NLR) for the patients in each Glasgow Outcome Scale Extended Pediatric Version score outcome category (GOS-E Peds; no or mild disability = 1–2; moderate or severe disability = 3–6; vegetative state or death = 7–8). Complete blood count (CBC) times are approximate time after the traumatic head injury. The whiskers represent a single standard deviation above and below the mean.

the circulation into the brain at the site of injury [9,17,22]. The association of high NLR with worse outcomes suggests that using the NLR to measure the degree of neutrophil response to the original TBI and potential sequelae is a first step in understanding the pathophysiology of this process and targeting potential therapeutic interventions.

These findings suggest that the level of neutrophil response in the first few days after TBI may be predictive of eventual outcome. NLR is an objective measure and is easier to document than many clinical measures such as GCS, PTA, and LOC, which are subjective in nature and can be more difficult to ascertain in a pediatric population. This makes NLR an easily obtained clinical marker that may assist prognosis of TBI and that could potentially become a target for treatment if pharmacological regulation of neuroinflammation is shown to improve TBI outcomes. Considering the lack of association between NLR and GCS in the present study, it is possible that NLR could be used in conjunction with GCS to improve upon prognostication, although further investigations would be needed, as this finding may instead be due to study limitations.

The limitations of this study include the retrospective nature of data collection and variations in the times at which laboratory values were obtained. Because this was a retrospective review, the times of injury were also best approximations based on the data available in the electronic medical record. The information on LOC and PTA were obtained from the Trauma One registry. The information for this registry is entered by trained registrars so it can be used by multiple institutions. However, there is always possible variability between physicians in determining LOC and PTA. Because of the lack of a formal outcome evaluation by one physician at a specific time point for each patient, there is variability in the outcome scores and this is a limitation that may skew results. We have included ages 0–18 in order to power this study, but a sub analysis of different age groups due to the difference in plasticity in children would be important to analyze in future studies. The relationship between NLR and TBI outcome could also be subject to confounding variables, such as damage to other organs and complications such as pneumonia. However, these variables reflect the heterogeneous nature of traumatic injury, and exclusion of patient data based on these variables could limit the generalizability of these findings. Additionally, because NLR values after hospital discharge were not available for most patients, these findings are limited to patients who remained in the hospital more than 12 h post-injury and may not apply to patients with uncomplicated TBI. However, an association between higher NLR and pathology on head CT among patients with isolated mild TBI has been observed previously [35]. Length of hospital stay was reported but not included in the analysis because, particularly in a pediatric population, some patients' length of stay may be influenced by factors unrelated to severity of injury (for example, if non-accidental trauma is suspected). The lack of statistical significance at later time points may be in part due to the decreased numbers of CBCs with differentials obtained at later time points or could reflect the pathogenic mechanism of neutrophil-mediated secondary tissue injury in the acute response to TBI, rather than an ongoing chronic neuroinflammation; thus, further investigation with larger studies is warranted.

5. Conclusion

In this study, higher peripheral blood NLRs 1 and 2 days after injury were associated with worse outcomes among pediatric TBI patients. NLR may be a useful measure to consider in TBI prognostication, as well as a potential target for therapeutic intervention. Because NLR is easy to calculate based on common, cost-effective blood tests, further studies are warranted that use larger prospective trials, apply more precise collection times, and include other inflammatory markers and clinical variables such as vomiting and headaches.

Disclosure

The authors report no conflict of interest concerning the materials or methods used in this study or the findings specified in this paper.

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Previous presentation

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CRediT authorship contribution statement

Rebekah Kimball: Methodology, Formal analysis, Investigation, Writing - original draft, Visualization. **Elad Shachar:** Investigation, Resources. **Stephanie Eyerly-Webb:** Methodology, Validation, Writing - review & editing, Visualization. **Daxa M. Patel:** Supervision, Writing - review & editing. **Heather Spader:** Supervision, Methodology, Conceptualization, Resources, Writing - original draft, Project administration.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.clineuro.2020.105772>.

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